

Transfer Rates -- Guided Notes ANSWER KEY

Unit tp-1.6 | CompTIA Network+ N10-009 Obj. FC0-U71 1.3 | N10-008 FC0-U71 1.3

For instructor use / distribute after students complete the notes.

Question 1

Network and connection speeds are measured in _____ per second (lowercase b), while file sizes are measured in _____ (uppercase B). To convert from Mbps to MB/s, you divide by _____. A 100 Mbps internet connection transfers data at _____ MB/s.

Answer: Megabits (Mbps); megabytes (MB); 8; 12.5 MB/s.

Question 2

USB 2.0 has a maximum transfer speed of _____ Mbps. USB 3.0 has a maximum of _____ Gbps. USB 3.0 is approximately _____ times faster than USB 2.0. USB 3.0 ports are typically identified by their _____ color inside the port.

Answer: 480 Mbps; 5 Gbps; approximately 10 times faster; blue.

Question 3

Calculate how long it would take to transfer a 1 GB file over a USB 2.0 connection at a sustained 480 Mbps:

Convert speed to MB/s: $480 \text{ Mbps} / 8 = \text{_____ MB/s}$

1 GB = _____ MB

Time = $1,024 / \text{_____} = \text{approximately _____ seconds}$

Answer: 60 MB/s; 1,024 MB; $1,024 / 60 = \text{approximately 17 seconds}$.

Question 4

_____ refers to the maximum theoretical data transfer capacity of a connection, usually measured in Mbps or Gbps. _____ is the actual measured data transfer rate achieved in real conditions, which is always _____ (greater/less) than the theoretical maximum due to overhead and interference.

Answer: Bandwidth; throughput; less.

Question 5

_____ is the delay between when a signal is sent and when it is received, measured in milliseconds (ms). It does not affect _____ (how much data per second) but affects how responsive a connection feels. High latency causes noticeable delays in _____ gaming, video calls, and remote desktop sessions.

Answer: Latency; bandwidth (throughput); online.

Question 6

An ISP advertises an internet plan as 500 Mbps download. A user downloading a file notices the download progress bar shows around 60 MB/s. Is this accurate or is there a problem? Show the conversion:

500 Mbps / 8 = _____ MB/s

60 MB/s is _____ (above/below/at) the theoretical maximum.

Answer: $500 / 8 = 62.5$ MB/s. 60 MB/s is slightly below the theoretical maximum -- this is normal due to protocol overhead. No problem.

Question 7

Mbps stands for _____ per second. MBps stands for _____ per second. The difference is _____ times larger. Internet service providers advertise in _____ (Mbps/MBps) while download managers typically display progress in _____ (Mbps/MBps).

Answer: Megabits; megabytes; 8 times; Mbps; MBps (MB/s).

Question 8 -- Real World Application

REAL WORLD: A student is downloading a 4 GB game on a home internet connection advertised at 200 Mbps. Their download client shows the download speed fluctuating between 15 MB/s and 20 MB/s, averaging about 18 MB/s. They expect much faster speeds based on the 200 Mbps advertisement. Explain why actual speed is lower and calculate the expected download time at 18 MB/s.

Answer: The 200 Mbps advertisement is the maximum theoretical bandwidth. Converting: $200 \text{ Mbps} / 8 = 25 \text{ MB/s}$ theoretical maximum. Actual speeds are lower due to: network congestion at the ISP, protocol overhead (TCP headers, acknowledgments), Wi-Fi signal strength or interference if not on wired Ethernet, and server-side upload limits. 18 MB/s is approximately 72% of the theoretical max -- a reasonable real-world figure. Download time: $4 \text{ GB} = 4,096 \text{ MB}$. $4,096 / 18 \text{ MB/s} = \text{approximately } 228 \text{ seconds}$, or about 3.8 minutes.